Flight & Airline Analytics Platform

1. Project Introduction

In the rapidly evolving aviation sector, operational efficiency and punctuality are paramount. This project aims to design and implement a scalable, high-performance data platform to analyze aviation operations across major airports (JFK, LGA, EWR). Beyond traditional historical reporting, the platform integrates real-time streaming and advanced Machine Learning to provide a holistic view of airport health and deliver proactive predictions for flight delays and congestion.

2. Technology Stack & Architecture

The system is built on a hybrid architecture that seamlessly integrates two primary data pathways:

A. Streaming & Predictive Pipeline

- **Data Sources:** Real-time flight data is ingested via the **OpenSky Network API**, while live weather conditions are pulled from the **Open-Meteo API**.
- **Ingestion: Apache Kafka** serves as the central message broker, managing high-throughput data streams with low latency.
- **Real-time Processing & ML: PySpark Structured Streaming** processes incoming data on the fly. Within this layer, an **XGBoost Classifier**—the platform's machine learning core—analyzes combined flight and weather features to predict the probability of flight delays in real-time.
- **Storage & Visualization:** Processed data and predictions are stored in **TimescaleDB** (a time-series database) and visualized through live **Grafana** dashboards, which trigger automated alerts for predicted delays or capacity breaches.

B. Batch Processing Pipeline

- **Orchestration: Apache Airflow** manages and schedules the end-to-end workflow, ensuring reliable task execution.

- **Raw Storage:** Historical datasets, including flight records and weather archives, are stored in **Amazon S3** as a robust Data Lake.
- **Data Processing: Apache Spark** performs heavy-duty data cleaning and complex transformations to prepare data for analytical modeling.
- **Data Warehouse & BI:** Curated data is loaded into **Snowflake**, providing a high-performance analytical layer connected to **Power BI** for detailed executive reporting and trend analysis.

3. Target Audience & Business Impact

This platform provides actionable insights for key stakeholders in the aviation ecosystem:

- **Airline Operations Teams:** Monitor fleet performance and leverage XGBoost predictions to preemptively adjust schedules and mitigate the financial impact of delays.
- **Airport Management:** Detect peak congestion periods and optimize gate and runway allocation to improve overall throughput and passenger flow.
- **Air Traffic Control (ATC):** Access real-time monitoring of aircraft movements in relation to shifting weather patterns for enhanced situational awareness.
- **Business Analysts & Decision Makers:** Utilize Power BI dashboards to identify seasonal trends, evaluate airport performance, and support strategic route planning.
- **Travel Technology Platforms:** Integrate real-time delay predictions into consumer-facing applications to enhance passenger experience through early notifications.

4. Core Use Cases

1. **Flight Delay Analysis:** Analyze flight delays across major airports to identify patterns and key factors affecting flight performance. Key outputs include delay rate per airport, average delay duration, delay trends, and airport comparisons.
2. **Weather Impact on Delays:** Evaluate how different weather conditions affect flight delays and operational performance. Key outputs include correlation between weather and delays, identification of high-risk weather scenarios, and delay distribution.

3. **Airport Congestion Analysis:** Analyze airport traffic and detect congestion levels that contribute to delays. Key outputs include the number of flights per hour/day, detection of peak periods, and identification of overloaded airports.

4. **Real-Time Delay Prediction:** Predict the likelihood of flight delays in real-time by feeding live flight and weather data into a trained machine learning model.

5. Team Information

Team Number: 19

ID	Team Member
1	Ahmed Refat Abd Elmoaty
2	Yasmein Assad Taher
3	Shaimaa Mohamed Tony
4	Abdelaal Ahmed Mohamed
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